DSA REFERENCE

2D Dynamic Programming Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Represent progress along two dimensions—positions, capacities, or strings—and reuse already solved cells.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / DYNAMIC-PROGRAMMING-2D

DSA

2D Dynamic Programming Failure Modes

CORE IDEA

Represent progress along two dimensions—positions, capacities, or strings—and reuse already solved cells.

Recognition cues

+ two sequences
+ grid paths
+ capacity plus item index
+ edit or alignment operations

Implementation sequence

+ name both axes
+ derive predecessor cells
+ initialize first row and column
+ verify traversal order
+ separate reconstruction from score

Invariant

Each cell answers the same question for a precisely smaller pair of dimensions.

Worked reasoning

Longest common subsequence compares final characters; equal characters extend the diagonal, otherwise take the better neighbor.

Complexity

Commonly O(rows × columns) time and space; rolling rows can reduce memory when reconstruction is unnecessary.

Failure modes

- [ ] off-by-one indexing between strings and table
- [ ] uninitialized boundaries
- [ ] reading future cells
- [ ] compressing before validating dependencies

Practice ladder

2D Dynamic Programming Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Unique Paths**	State the cue, invariant, edge case, and proof before implementation.
2	**Longest Common Subsequence**	State the cue, invariant, edge case, and proof before implementation.
3	**Edit Distance**	State the cue, invariant, edge case, and proof before implementation.
4	**Coin Change II**	State the cue, invariant, edge case, and proof before implementation.
5	**Distinct Subsequences**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.